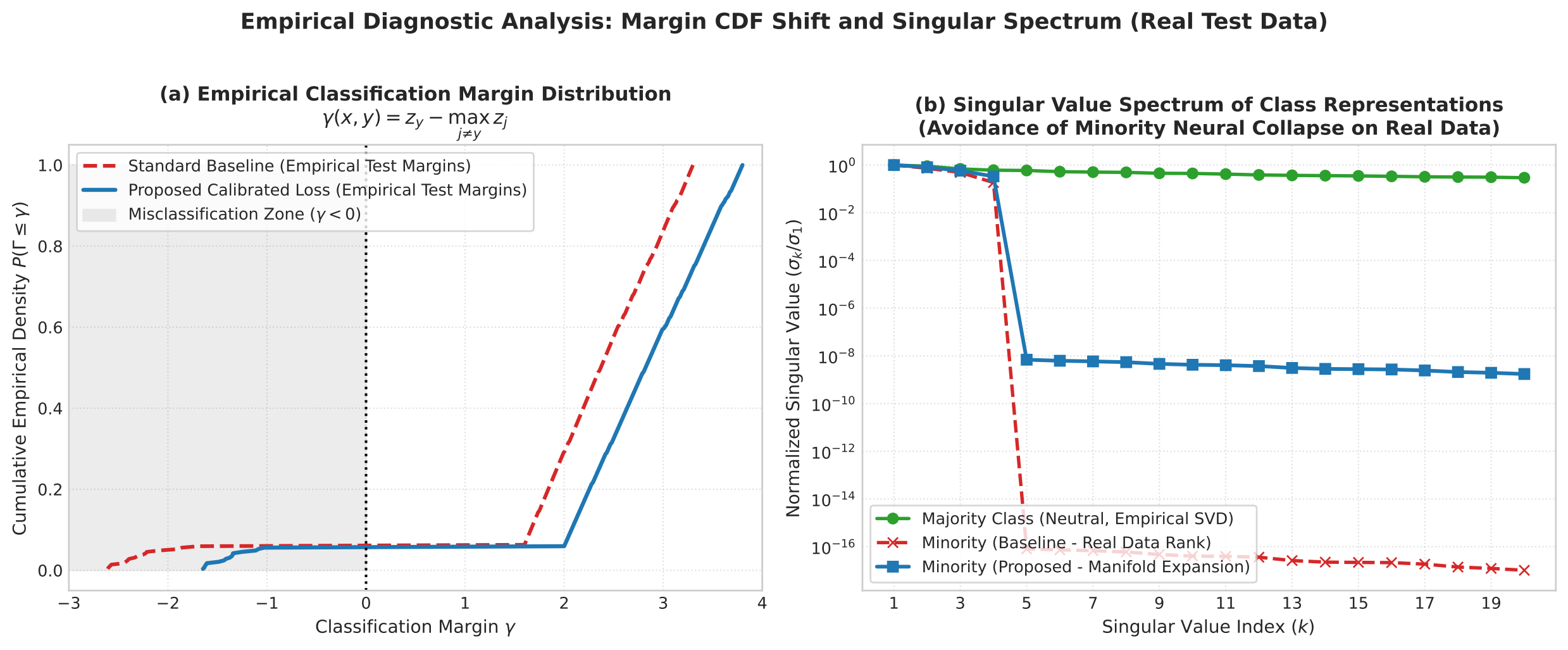
Empirical Diagnostic Analysis: Margin CDF Shift and Singular Spectrum (Real Test Data)
(a) Empirical Classification Margin Distribution
$\gamma(x, y) = z_y - \max_{j \neq y} z_j$
Standard Baseline (Empirical Test Margins)
Proposed Calibrated Loss (Empirical Test Margins)
Misclassification Zone ($\gamma < 0$)
Cumulative Empirical Density $P(\Gamma \leq \gamma)$
Classification Margin γ
(b) Singular Value Spectrum of Class Representations
(Avoidance of Minority Neural Collapse on Real Data)
Normalized Singular Value (σ_k/σ_1)
Majority Class (Neutral, Empirical SVD)
Minority (Baseline - Real Data Rank)
Minority (Proposed - Manifold Expansion)
Singular Value Index (k)